

Science - K-12 Level Assessment 6

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

Most of the oxygen in the atmosphere is made by algae. Where do most of the algae live?

A. coastal ocean

B. deep ocean

C. ponds

D. lakes

Question 2

(2 marks)

In the design process, a trade-off occurs when a problem is solved but a feature is sacrificed. Which of the following is an example of a trade-off?

A. A car's gas mileage is increased and the engine has less power.

B. An airplane uses a more efficient engine and has higher performance.

C. A sports drink's taste is improved and has the same nutritional content.

D. A computer company upgrades the hardware and the price remains unchanged.

Question 3

(2 marks)

Lithospheric plates have moved to form land formations on the Earth's surface. Which land formation occurs as a result of the plates folding upon one another?

A. an earthquake

B. a mountain range

C. a river

D. a volcano

Question 4

(2 marks)

Which changes will most likely have a negative effect on an ecosystem?

A. building a road

B. planting a tree

C. adding a freshwater source

D. creating a nature sanctuary

Question 5

(2 marks)

Which geologic process most likely caused the formation of the Mount St. Helens Volcano?

A. converging boundaries

B. diverging boundaries

C. transform faults

D. rift zones

True / False

Question 1

(2 marks)

Comparing an experiment's results to a control group can help disprove unfounded hypotheses, such as the transformation of hair into worms.

True

False

Question 2

(2 marks)

Aphids are first-level consumers that feed on plants in a food web.

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True

False

Question 3

(2 marks)

Light experiences the same amount of refraction when passing through a telescope lens as it does through a tinted window.

True

False

Question 4

(2 marks)

A solution with a pH of 2 can be raised above 7 by adding a base, which neutralizes the acidity.

True

False

Question 5

(2 marks)

The sun would be enormous when compared to an average star in size.

True

False

Long Form

Question 1

(10 marks)

Discuss the forces acting on a fork as it falls off the edge of a table, explaining how gravity influences its movement toward the floor.

Type your answer here...



Question 2

(10 marks)

Discuss the importance of friction in bicycle ramps and analyze which materials could be used to enhance friction, providing examples and justifications for your choices.

Type your answer here...



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